

Problem Set 3 — Intractability & Heuristics

VCE Algorithmics (HESS) · Unit 4 AOS2 · Week 6

Due: start of Week 7. Question E1 feeds directly into Memo 03.

Name: _____

Due: Monday 24 August

Section A — Classes and forms

A1. For each problem, write its decision form:

- a.) Find the shortest tour visiting all n warehouses.
- b.) Pack maximum-value equipment into a 20 kg drone payload.
- c.) Timetable 12 classes into the fewest rooms with no clashes.

A2. Define P and NP in one sentence each, and explain why $P \subseteq NP$.

A3. "TSP is in NP" — state exactly what must be checked in polynomial time for this claim, and give the complexity of that check.

Section B — Reductions

B1. Hamiltonian circuit reduces to TSP-decision (as shown in class: weight 1 for existing edges, weight 2 otherwise, threshold $|V|$). Explain why a "yes" answer to the TSP instance implies a Hamiltonian circuit exists, and vice versa.

B2. A student says: "Problem A reduces to problem B, and A is NP-complete, so B is easy." Diagnose the error and state the correct conclusion about B.

Section C — Exact techniques

C1. Naive recursive Fibonacci is $\Theta(\phi^n)$; memoised Fibonacci is $\Theta(n)$. State the two properties of the problem that make dynamic programming applicable, and identify where each is visible in Fibonacci.

C2. The 0/1 knapsack DP runs in $O(n \cdot W)$ for n items and capacity W . Explain why this does *not* contradict knapsack's NP-hardness. (*Hint: how does W relate to the length of the input?*)

C3. For the backtracking template from class, explain what "pruning" means and why the worst case remains exponential despite it.

Section D — Hill climbing

D1. Define, for TSP with the 2-opt neighbourhood: the solution representation, the neighbourhood of a solution, and the objective function.

D2. Explain the local optimum problem with a one-sentence landscape metaphor, and give two techniques to

mitigate it.

D3. "Hill climbing solved my TSP instance." Give two reasons this claim is imprecise, referring to guarantees.

Section E — SAT bridge

E1. *Memo 03 rehearsal.* Consider the bottleneck module you named in Memo 02.

- a.) Does your underlying problem (or a component of it) resemble an NP-hard problem from the recognition table? Which one, and why?
- b.) If yes: propose a heuristic (representation, neighbourhood, objective) and state the trade-off sentence. If no: propose a better exact approach and its bound.
- c.) State one constraint from your problem description that your improved approach must still respect.
